From Classroom/Integrated School Programs 7 in Top 20, 23 in Top 100, 54 in Top 300, 106 in Top 500 All India Ranks & 2314 Students from Classroom /Integrated School Programs & 3723 Students from All Programs have been Awarded a Rank in JEE (Advanced), 2013

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JEE(Advanced)-2014

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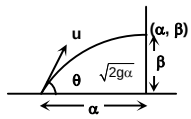
CRT –II
(Paper-2)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	A	B	D
2.	B	D	B
3.	C	A	B
4.	C	A	B
5.	D	D	B
6.	A	D	B
7.	A	A	B
8.	A	C	C
9.	A	D	B
10.	A	D	A
11.	D	A	B
12.	A	B	A
13.	B	D	A
14.	A	B	C
15.	B, D	B, C, D	A, C
16.	B, C	A, B	A, D
17.	A, C	A, B	A, B, C
18.	B, C, D	A, B, D	B, D
19.	B, D	A, D	A, D
20.	A, B, C, D	A, C, D	A, C, D

Physics

PART – I

2. $\beta = \alpha \tan\theta - \frac{1}{2} g \frac{\alpha^2}{u^2 \cos^2\theta}$



$\beta = \alpha \tan\theta - \frac{1}{2} g \frac{\alpha^2}{2g\alpha}\left(1+\tan^2\theta\right)$

$\tan^2\theta - 4\tan\theta + \left(1+\frac{4\beta}{\alpha}\right) = 0$

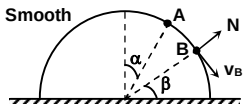
If particle will not hit the target.

$(b^2 - 4ac) < 0$

$16 - 4\left(1+\frac{4\beta}{\alpha}\right) < 0$

$4\beta > 3\alpha$

3. At B, $mg \sin\beta = \frac{mv_B^2}{r}$ . . . (1)



Using energy considerations

$\frac{1}{2}mv_B^2 = mgr(\cos\alpha - \sin\beta)$ . . . (2)

From (1) and (2)

$mg \sin\beta = 2mg(\cos\alpha - \sin\beta)$

$2\cos\alpha = 3\sin\beta$

7. $i = \sqrt{5}A$

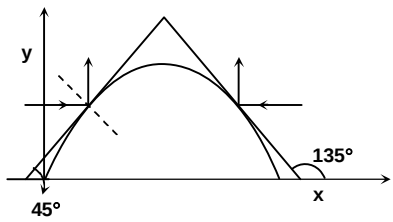
$\frac{q_m^2}{2C} = \frac{q^2}{2C} + \frac{1}{2}Li^2 \Rightarrow q_{max} = 6\ C$

8. Time period does not depend on amplitude SHM and both particle will exchange velocity at every collision.

15. The slope of curve at such a point will be ± 1

$\therefore \quad \frac{dy}{dx} = 2\cos\left(\frac{\pi x}{L}\right) = \pm 1$

$\Rightarrow \quad x = \frac{L}{3}$ or $\frac{2L}{3}$



16. $P_1 = m\times 16 + 2m\times 0 = 16m$

$P_f = mv_A' + 2mv_B'$

$\therefore \ \frac{1}{2}\times 2mv_B^2 = 2mgh$

$\Rightarrow v_B = 10\ m/s$

$16\ m = mv_A' + 2m\times 10$

$\Rightarrow v_A' = -4m/s$

$e = \frac{v_B' - v_A'}{v_A - v_B} = \frac{10-(-4)}{16-0} = \frac{14}{16} = \frac{7}{8}$

17. If v be the velocity at mean position in two cars then $\frac{1}{2}mv^2 = \frac{1}{2}KA_1^2 = \frac{1}{2}KA_2^2$

and $T_1 = T_2 = 2\pi\sqrt{\frac{m}{K}}$

$A_1 = A_2$

18. Initially potential and kinetic both energies zero and from conservation of mechanical energy total energy of the two object zero
Further, decrease in P.E. = increase in K.E.

$$\frac{G(m)(4m)}{r} = \frac{1}{2}\mu v_r^2$$

$$v_r = \sqrt{\frac{10Gm}{r}}$$

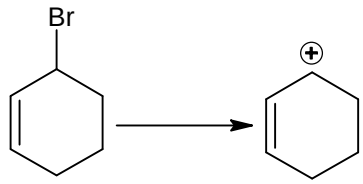
Total K.E. $= \frac{G(m)(4m)}{r} = \frac{4Gm^2}{r}$

Chemistry

PART – II

SECTION – A

1. XeO_6^{4-} (perxenate ion) because oxidation state of Xe is +8 and it is highly unstable compound.

2. 

Due to formation of resonance stabilized carbocation.

3. $\pi = CRT, So, C = \frac{\pi}{RT} = \frac{293.4}{8.31 \times 293} = 0.1 \text{ mol/L}$

If 100 ml contains 6.33 g of haematin.

$\therefore$ 1 L will contain 63.3 g of haematin.

Molecular mass of haematin = 633 u

Estimating the empirical formula of compound

$$C : H : N : O : F = \frac{64.6}{12} : \frac{5.2}{1} : \frac{8.8}{14} : \frac{12.6}{16} : \frac{8.8}{56}$$

$\approx 34 : 33 : 4 : 5 : 1$

So, empirical formula is $C_{34}H_{33}N_4O_5Fe$, whose mass corresponds to the molecular mass.

4. $E^o_{Zn^{2+}/ZnY^{2-}/Y^{4-}} = E^o_{Zn^{2+}/Zn^+} + \frac{0.059}{2}\log K$

$$K_f = \frac{\left[ZnY^{2-}\right]}{\left[Y^{4-}\right]} \Rightarrow K = \frac{1}{K_f}$$

$$E^o_{Zn^{2+}/ZnY^{2-}/Y^{4-}} = -0.76 + \frac{0.059}{2}\log_{10}\frac{1}{3.2 \times 10^{16}}$$

= - 1.25 V

5. $\Delta T_f = 0.5K = iK_f m$

$$\therefore i = \frac{\Delta T_f}{K_f m} = \frac{0.5}{1.86 \times 0.263} = 1.022$$

$HA \longrightarrow$	H^+	+	A^-
1	0		0
$1-\alpha$	α		α

$i = 1 + \alpha \Rightarrow \alpha = 0.022$

$K_a = C\alpha^2 = 0.2 \times (0.022)^2 = 9.6 \times 10^{-5}$

$\Rightarrow pK_a = 4.0177$

On adding 0.25 M NaA, buffer is formed.

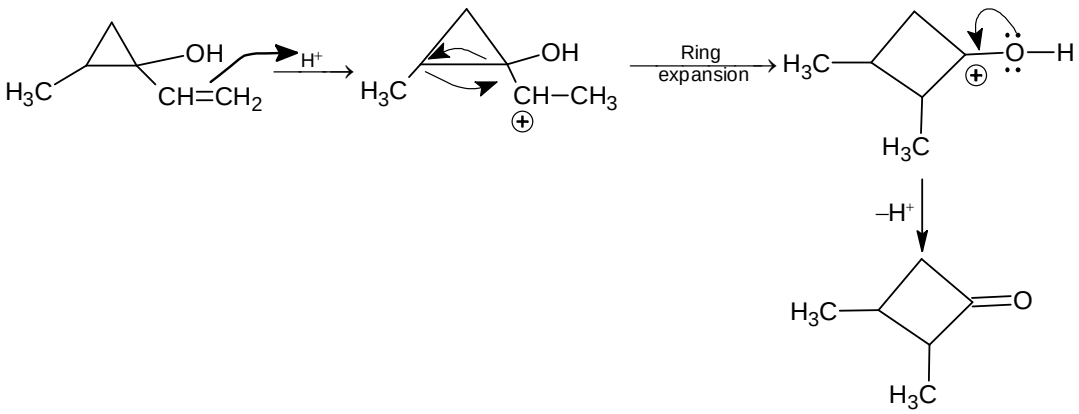
$$\therefore pH = pK_a + \log_{10}\frac{[Salt]}{[Acid]} = 4.0177 + 0.0969$$

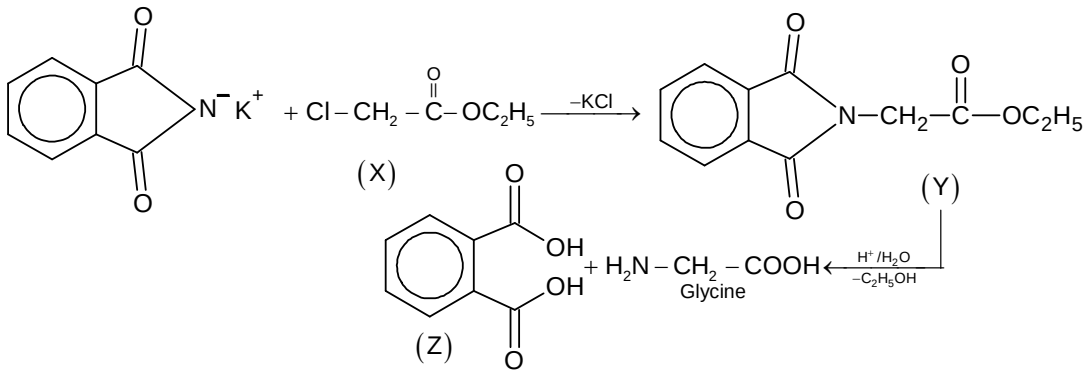
= 4.1146

6. $Na_2B_4O_7 + 7H_2O \longrightarrow \underbrace{\underbrace{2NaOH}_{\text{Strong base}} + \underbrace{4H_3BO_3}_{\text{Weak acid}}}_{\text{Alkaline}}$

$$Na_2B_4O_7 \xrightarrow{\Delta} \underbrace{2NaBO_2 + B_2O_3}_{\text{Glassy bead}}$$

$$\underset{(\text{White crystals})}{Na_2B_4O_7} + H_2SO_4 + 5H_2O \longrightarrow Na_2SO_4 + 4H_3BO_3$$

7. 

8. 

9. $\underset{(A)}{3M} + N_2 \xrightarrow{\text{Burning}} \underset{(B)}{M_3N_2}$

$$\underset{(B)}{M_3N_2} + 6H_2O \longrightarrow \underset{(C)}{3M(OH)_2} + \underset{(D)}{2NH_3}$$

All statement with respect to NH_3 are correct.

10. Ba and Ca quite readily liberate hydrogen.
In cold water Mg decomposes water only on heating. So, element (A) may be Ca and Ba not Mg.
$\underset{(\text{Lime water})}{Ca(OH)_2}$ & $\underset{(\text{Baryta water})}{Ba(OH)_2}$, both givess milkiness with CO_2.

11. $$2.303\log\frac{P_2}{P_1} = \frac{\Delta H}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

$\Delta H = 11.99\ kJmol^{-1}$

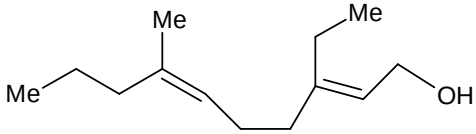
12 $K_b = \dfrac{RT^2M}{1000\ \Delta H} = \dfrac{8.314\times(353)^2\times100}{1000\times11.99\times10^3} = 8.64\ K\,kg\,mol^{-1}$

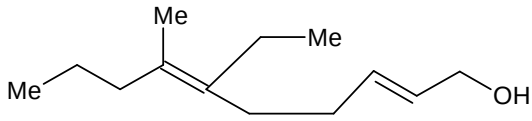
$\Delta T_b = K_b(1+\alpha)m$

$1 = 8.64\times(1+\alpha)\dfrac{1}{100}\times\dfrac{1000}{100}$

$\alpha = 0.158$

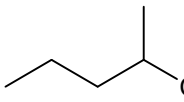
13. Two structural isomers of (A) are:



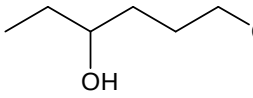
&


There are four geometrical isomers for each strucutre, i.e. total number of stereoiosmers = 8.

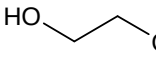
14. Convert aldehyde and ketone to alcohol

(1) 

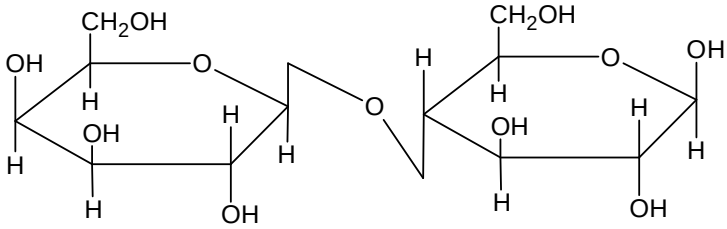
$(5C-)$

(2) 

$(6C-)$

(3) 

$(2C-)$

15. 

16. (A) 

$\mu_{NH_3} > \mu_{NF_3}$

(B) (I) is 3° allylic with extended conjugation.
(II) is 3° allylic, (III) is 3° free radical.

(C) Correct order is: $RS^- < CH_3O^- < OH^-$ (Basic strength)

(D) Correct order is: $I^- < Br^- < Cl^- < F^-$ (Basic strength)

17. (A) $NH_4NO_3 + NaOH \longrightarrow NH_3 + NaNO_3 + H_2O$

$7NaOH + NaNO_3 + 4Zn \longrightarrow 4Na_2ZnO_2 + NH_3 + 2H_2O$

(B) $NH_4NO_2 + NaOH \longrightarrow NaNO_2 + NH_3 + H_2O$

$3Zn + 5NaOH + NaNO_3 \longrightarrow 3Na_2ZnO_2 + NH_3 + H_2O$

18. $v_{rms} = \sqrt{\dfrac{3RT}{M}} = \sqrt{\dfrac{3PV}{M}} = \sqrt{\dfrac{3P}{d}}$

19. In metal carbonyls, $M \underset{\sigma}{\overset{\pi}{\rightleftharpoons}} CO$, the C – O bond length increases compared to that in CO molecules due to synergic bonding between metal and carbonyl ligand.

Facial and meridenal isomers $[Ma_3b_3]^{n+}$ contains plane of symmetry.

20. $CH_3COO^- + H_2O \rightleftharpoons CH_3COOH + OH^-$

$0.1(1-h)$ $0.1h$ $0.1h$

$$K_h = \frac{(0.1h)(0.1h)}{0.1(1-h)} = 0.1h^2 \Rightarrow h = 7.48\times10^{-5}$$

$$\Rightarrow [OH^-] = Ch = 7.48\times10^{-5}\times10^{-1} = 7.48\times10^{-6}$$

$$\therefore [H^+] = \frac{K_W}{[OH^-]} = \frac{10^{-14}}{7.48\times10^{-5}} = 1.33\times10^{-9}$$

And pH = 8.8 approx.

Mathematics

PART – III

1. We express x in terms of a new variable y

$x-1=\frac{1}{y} \Rightarrow x=\frac{y+1}{y}$

$\Rightarrow \frac{6(y+1)^5}{y^5}+\frac{5(y+1)^3}{y^3}+\frac{(y+1)^2}{y^2}+\frac{4(y+1)}{y}-1=\frac{a}{y^5}+\frac{b}{y^4}+\frac{c}{y^3}+\frac{d}{y^2}+\frac{e}{y}+f$

Multiplying both side by y^6 we get $6y(y+1)^5 + 5y^3(y+1)^3 + y^4(y+1)^2 + 4y^5(y+1) - y^6 = ay + by^2 + cy^3 + dy^4 + ey^5 + fy^6$

Differentiating both side and then satisfying y = 1

Gives the value a + 2b + 3c + 4d + 5e + 6f = 910

2. $f(x)=\prod_{n=1}^{50}(x-n)^{n(51-n)}$

$\ln(f(x))=\sum_{n=1}^{50} n(51-n)\ln(x-n)$

Differentiating both side w.r.t. x

$\frac{f'(x)}{f(x)}=\sum_{n=1}^{50}\frac{n(51-n)}{x-n}$

$\Rightarrow \frac{f'(51)}{f(51)}=1275$

3. $a^2 = b^2 + c^2 - 2bc\cos A = (b-c)^2 + 2bc(1-\cos A)$

$\Delta = \frac{1}{2}bc\sin A \Rightarrow 2bc=\frac{4\Delta}{\sin A}$

$\Rightarrow a^2=(b-c)^2+\frac{4\Delta}{\sin A}(1-\cos A) = (b-c)^2 + 4\Delta\tan\frac{A}{2}$

Since A and Δ are fixed hence a is minimum when b = c

Hence $2bc = 2b^2 = \frac{4\Delta}{\sin A}$

4. $\frac{f(3x+3y)}{\sin(2x+3y)}=\frac{f(3x-3y)}{\sin(2x-2y)} \quad \forall\ x, y \in R$

$\Rightarrow f(x)=k\sin\left(\frac{2x}{3}\right)$

$\therefore\ f'(0)=\frac{2}{3}k=1 \Rightarrow k=\frac{3}{2}$

$\therefore\ f(x)=\frac{3}{2}\sin\left(\frac{2x}{3}\right) \Rightarrow f'\left(\frac{\pi}{2}\right)=\frac{3\sqrt{3}}{4}$

Equation of tangent is $y-\frac{3\sqrt{3}}{4}=\frac{1}{2}\left(x-\frac{\pi}{2}\right)$

5. $\sum_{1\le i<j\le m} a_i a_j = \frac{1}{2}\left\{\left(\sum_{i=1}^{m}a_i\right)^2-\sum_{i=1}^{m}a_i^2\right\}$

$$= \frac{1}{2}\left\{\left(\frac{a_1(1-r^m)}{1-r}\right)^2 - \frac{a_1^2(1-r^{2m})}{1-r^2}\right\}$$

$$= \frac{a_1^2}{2(1-r)}\left\{\frac{(1-r^m)^2}{1-r} - \frac{1-r^{2m}}{1+r}\right\}$$

$$= \frac{r}{1+r}\left\{\frac{a_1(1-r^{m-1})}{1-r}\right\}\left\{\frac{a_1(1-r^m)}{1-r}\right\}$$

$$= \frac{r}{1+r}S_{m-1}S_m$$

6. Use L'Hospital's Rule twice

7. $g(x) = \int \frac{f(x)dx}{x^2(x+1)^3} = \int\left(\frac{A}{x} + \frac{B}{x^2} + \frac{C}{x+1} + \frac{D}{(x+1)^2} + \frac{E}{(x+1)^3}\right)dx$

$= A\ln x - \frac{B}{x} + c\ln(1+x) - \frac{D}{1+x} - \frac{E}{2(x+1)^2}$

Since g(x) is a rational function, hence logarithmic function must be absent

$\Rightarrow A = C = 0$

$g(x) = \int \frac{B}{x^2} + \frac{D}{(x+1)^2} + \frac{E}{(x+1)^3}\,dx$

$\Rightarrow f(x) = (B + D)x^3 + (3B + D + E)x^2 + 3Bx + B$

$\Rightarrow B + D = 0$, $f(0) = 1$ gives $B = 1 \Rightarrow D = -1$

$\therefore f(x) = (2 + E)x^2 + 3x + 1$

$f'(x) = 2(2 + E)x + 3$

$f'(0) = 3$

8. Denoting the two curve by S_1 and S_2 the equation of curve S passing through intersection of S_1 and S_2 can be $S = S_1 + \lambda S_2 = 0$. Since S is circle

$\Rightarrow$ Coefficient of x^2 = Coefficient of y^2, Coefficient of $xy = 0$ we get $\lambda = 1$

$S = (a' + a)(x^2 + y^2) - 2(g' + g)x - 2(f' + f)y + 2c = 0$

The centre is $\left(\frac{g'+g}{a'+a}, \frac{f'+f}{a'+a}\right)$ which is P

$PA^2 + PB^2 + PC^2 = 3(\text{Radius})^2 = 3(PD)^2$

9. $\sin x = \sin\frac{(\pi x)}{180} \Rightarrow \sin x = \sin\left(\pi - \frac{\pi x}{180}\right)$ or $\sin\left(2\pi + \frac{\pi x}{180}\right)$

$x = \pi - \frac{\pi x}{180}$, $x = 2\pi + \frac{\pi x}{180}$

$x = \frac{180\pi}{180+\pi}$, $x = \frac{360\pi}{180-\pi}$

On comparing, m = 360, n = 180, p = 180, q = 180

$\Rightarrow (m + n + p + q) = 900$

10. $\lim_{x\to\infty} \frac{(A - C^2)x^2 + Bx}{x\left(\sqrt{A + \frac{B}{x}} + C\right)} = 2$

For existence of limit $A = C^2$

Hence, $C = \sqrt{A}$, $\frac{B}{\sqrt{A}+C} = 2$

$\Rightarrow \frac{B}{C} = 4$

For $C = -\sqrt{A}$ limit does not exist

11. The visible are of P(2, –6) is the area of ΔABP ΔABP has base AB of length 4 and its height is the distance from AB to point P. Which is 10, since AB is parallel to x–axis.

Thus area = $\frac{1}{2} \times b \times h = 20$

12. Similarly area = $\Delta QBC = \frac{1}{2} \times b \times h = \frac{1}{2} \times 4 \times 4 = 8$

13. $\text{LHD} = \lim\limits_{h \to 0} \left(\frac{-\sinh + \tanh + \cosh - 1}{\frac{2h^2 - \ln(2-h) - \tanh}{-h}} \right)$

$= \lim\limits_{h \to 0} \frac{\frac{\sinh}{h} - \frac{\tanh}{h} + \frac{1-\cosh}{h^2} \times h}{2h^2 - \ln(2-h) - \tanh} = 0$

$\text{RHD} = \lim\limits_{h \to 0} \frac{e^{h^2} - 1 - 0}{h} = h \times \frac{e^{h^2} - 1}{h} = 0$

$\Rightarrow L_1 \equiv y = 0, L_2 \equiv x = 0$

14. $\Delta = 2 \times \frac{1}{2} \left(\cot\left(\frac{\pi}{4} - \frac{\theta}{2} \right) + \cot \frac{\theta}{2} + 1 \right)$

$= 1 + \frac{2 \sin \frac{\pi}{4}}{2 \left(\sin\left(\frac{\pi}{4} - \frac{\theta}{2} \right) \right) \sin \frac{\theta}{2}}$

$= 1 + \frac{\sqrt{2}}{\cos\left(\theta - \frac{\pi}{4} \right) - \cos \frac{\pi}{4}}$

$\Delta_{\min} = 1 + \frac{\sqrt{2}}{1 - \frac{1}{\sqrt{2}}} = \left(1 + 2\left(\sqrt{2} + 1 \right) \right) = 3 + 2\sqrt{2}$

15. Four point A, B, C, D are coplanar if three vector $\overrightarrow{AB}$, $\overrightarrow{AC}$, $\overrightarrow{AD}$ are coplanar or STP of there vectors is 0

$\overrightarrow{AB} \cdot (\overrightarrow{AC} \times \overrightarrow{AD}) = 0$

$\Rightarrow \begin{vmatrix} 1 & 0 & 0 \\ 3 & f(t) & 1 \\ 2 & f'(t) & 2 \end{vmatrix} = 0 \quad \forall\, t \in R$

$\Rightarrow f'(t) = 2f(t)$

$\Rightarrow f(t) = ce^{2t}$

Similarly for $\vec{B}\left(\hat{i} + \hat{j} \right)$ differential equation will be $f'(t) - 2f(t) = -4$

On solving $f(x) = 2 + ce^{2x}$, $c \in R$

16. First use integrating by parts to show that

$u_n = \int_0^{\pi/2} \cos^{n-1}x \sin x \sin nx \, dx$

$= \int_0^{\pi/2} \cos^{n-1}x \cos(n-1)x - \int_0^{\pi/2} \cos^{n}x \cos nx \, dx$ (Use sin nx sin x = cos(n – 1)x – cos nx cos x)

$= u_{n-1} - u_n$

$\Rightarrow \frac{u_n}{u_{n-1}} = \frac{1}{2}$

We have, $u_1 = \frac{\pi}{4}$, $u_n = \frac{1}{2^{n-1}} \cdot \frac{\pi}{4} = \frac{\pi}{2^{n+1}}$

17. If three planes pass through the same line, their normal vector will be coplanar

$\Rightarrow \begin{vmatrix} -1 & \sin\alpha & \sin\beta \\ \sin\alpha & -1 & \sin\theta \\ \sin\beta & \sin\theta & -1 \end{vmatrix} = 0$

$\Rightarrow \sin^2\alpha + \sin^2\beta + \sin^2\theta + 2\sin\alpha\sin\beta\sin\theta = 1$ ..... (1)

Now the cross product $\hat{r}$ of normals of P_1, and P_2 will parallel to line

Direction ratio of $\hat{r} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & \sin\alpha & \sin\beta \\ \sin\alpha & -1 & \sin\theta \end{vmatrix}$ = {(sin α sin θ + sin β), (sin α sin β + sin θ), (1– sin² α)}

$\hat{r}$ is parallel to given line

$\Rightarrow \frac{\sin\alpha\sin\theta + \sin\beta}{\cos\theta} = \frac{\sin\alpha\sin\beta + \sin\theta}{\cos\beta} = \frac{1-\sin^2\alpha}{\cos\alpha} = \lambda$ ..... (2)

Using (1) and (2) we get

$\lambda \cos\alpha = \lambda \sin(\beta + \theta) \Rightarrow \sin(\beta + \theta) = \sin\left(\frac{\pi}{2} - \alpha\right) = \alpha + \beta + \theta = 2n\pi + \frac{\pi}{2}$ $(n \in z)$

18. Let the circle be $x^2 + y^2 = r^2$, and the five points be $P_i = (r\cos\theta_i, r\sin\theta_i)$, i = 1, 2, 3, 4, 5
Let (h_i, k_i) represent the centre of that rectangular hyperbola which passes through the four points P_j($j \neq i$) let (h_1, k_1) is centre of hyperbola which passes through P_2, P_3, P_4, P_5.

Now $\frac{h_1}{2} = \frac{r}{4}(\cos\theta_2 + \cos\theta_3 + \cos\theta_4 + \cos\theta_5)$, $\frac{k_1}{2} = \frac{r}{4}(\sin\theta_2 + \sin\theta_3 + \sin\theta_4 + \sin\theta_5)$

(As we know if a circle and rectangular hyperbola intersect at four point then their means is mid–point of centre of two curves)

$2h_1 = r\sum_{i=1}^{5}\cos\theta_i - r\cos\theta_1$

$\Rightarrow 2k_1 = r\sum_{i=1}^{5}\sin\theta_i - r\sin\theta_1$

However, r, $\left(\lambda = \sum_{i=1}^{5}\cos\theta_i\right)$ and $\left(\gamma = \sum_{i=1}^{5}\sin\theta_i\right)$ are fixed

$\Rightarrow 2h_1 = \lambda - r\cos\theta_1$, $2k_1 = \gamma - r\sin\theta$

$\Rightarrow \left(h_1 - \frac{\lambda}{2}\right)^2 + \left(k_1 - \frac{\gamma}{2}\right)^2 = \left(\frac{r}{2}\right)^2$

This equation is satisfied by all (h_i, k_i), i = 1, 2, 3, 4, 5

19. Divided the given equation by ax^4, and let $\frac{y}{x} = m$

Equation become $m^4 + \frac{b}{a}m^3 + \frac{c}{a}m^2 + \frac{d}{a}m + 1 = 0$ ..... (1)

Assuming that four slopes $m_1, m_2, m_3, m_4 \Rightarrow m_1 \cdot m_2 \cdot m_3 \cdot m_4 = 1$
Now it is given pair of these lines, bisect the angle between the other pair. So let, $m_1m_2 = -1$
$m_3m_4 = -1$ which means other two lines are also perpendicular
Equation (1) can be written as

$(m - m_1)(m - m_2)(m - m_3)(m - m_4) = (m - m_1)\left(m + \frac{1}{m_1}\right)(m - m_3)\left(m + \frac{1}{m_3}\right)$

$= (m^2 + \lambda m - 1)(m^2 + \mu m - 1)$ ..... (2)

Where $\lambda = \frac{1 - m_1^2}{m_1} \Rightarrow \mu = \frac{1 - m_3^2}{m_3}$

Comparing the coefficient (1) and (2)

$\lambda + \mu = \frac{b}{a}$, $\lambda\mu - 2 = \frac{c}{a}$, $\lambda + \mu = -\frac{d}{a}$ ..... (3)

Thus, $\frac{b}{a} = -\frac{d}{a} \Rightarrow b + d = 0$

Also, we have to use angle bisector fact

$$\left|\frac{m_1 - m_3}{1 + m_1m_3}\right| = \left|\frac{m_3 + \frac{1}{m_1}}{1 - \frac{m_3}{m_1}}\right| \Rightarrow (m_1 - m_3)^2 = (1 + m_1m_3)^2 \quad \text{..... (4)}$$

From (3), $\lambda\mu - 2 = \frac{c}{a} \Rightarrow \left(\frac{1 - m_1^2}{m_1}\right)\left(\frac{1 - m_3^2}{m_3}\right) - 2 = \frac{c}{a}$ ..... (5)

Using (4) and (5) we obtain $(c + 6a) = 0$

20. Since $P_k = e^{i\theta_k}$, $\theta_k = \frac{(2k)\pi}{n}$

Let $A_k = x_k + iy_k = a\cos\theta_k + ib\sin\theta_k$

$\Rightarrow \frac{x_k}{a} = \cos\theta_k$, $\frac{y_k}{b} = \sin\theta_k$, which implies that $\frac{x_k^2}{a^2} + \frac{y_k^2}{b^2} = 1$, A_k lies on ellipse

$$\Rightarrow \sum_{k=1}^{n} A_kS = \sum_{k=1}^{n}(a - ex_k) = na - e\sum_{k=1}^{n} x_k = na$$

$$\Rightarrow \sum_{k=1}^{n}(A_kS)^2 = \sum_{k=1}^{n}(a - ex_k)^2 = na^2 - 2ae\sum_{k=1}^{n} x_k + e^2\sum_{k=1}^{n} x_k^2$$

$$= na^2 + a^2e^2\sum_{k=1}^{n}\cos^2\theta_k = na^2 + \frac{na^2e^2}{2} + \frac{1}{2}a^2e^2\sum_{k=1}^{n}\cos 2\theta_k$$

$$= na^2 + \frac{na^2}{2}\left(1 - \frac{b^2}{a^2}\right) = \frac{n}{2}(3a^2 - b^2)$$

$(\because \sum_{k=1}^{n}\cos 2\theta_k = 0$ sum of roots of $z^{n/2} - 1 = 1)$